$$x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix} x, y \in R^n ; (x, y) = \sum_{i=1}^{n} x_i y_i$$

$(\cdot, \cdot) : R^n \, x \, R^n \to R$
(PS1) $(x, x) \geq 0; (x, x) = 0 \Leftrightarrow x = 0$
(PS2) $(x, y) = (y, x)$
(PS3) $(\alpha x_1 + \beta x_2, y) = \alpha(x_1, y) + \beta(x_2, y)$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_m \end{bmatrix} y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$$

$$x \cdot y^T = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_n \\ \dots & \dots & \dots & \dots \\ x_m y_1 & x_m y_2 & \dots & x_m y_n \end{bmatrix}; \text{rang}(x \cdot y^T) = 1$$

$$x_1 + x_2 + x_3 = 3$$
$$3x_1 + x_2 + x_3 = 5$$
$$x_1 + 4x_2 - x_3 = 4$$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{22}x_2 + \dots + a_{1n}x_n &= b_1 \\ \dots & \\ a_{\text{nn}}x_n &= b_n \end{aligned}$$

$$x_n = \frac{b_n}{a_{\text{nn}}}$$

$$x_k = \frac{1}{a_{\text{kk}}} \left(b_k - \sum_{j=k+1}^{n} a_{\text{kj_}} x_j \right)$$

```
function x = backsubst(A,b)
  n = length(b);
  x = zeros(n, 1);
  for k = n:-1:1
    x(k) = (b(k) - A(k, k + 1 :n) * x(k + 1:n)) / A(k, k);
  end
end

function x = forwardsubst(A,b)
  n = length(b);
```

```
  x = zeros(n, 1);
  for k = 1:n
    x(k) = (b(k) - A(k, 1:k) * x(1:k)) / A(k, k);
  end
end
```

$$\overline{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 3 & 1 & 1 & 5 \\ 1 & 4 & -1 & 4 \end{bmatrix}$$

$1^\circ\ (E_i) \leftrightarrow (E_j)$

$2^\circ\ \alpha(E_i) \leftrightarrow (E_j)$

$3^\circ\ \alpha(E_i) + (E_j) \leftrightarrow (E_j)$

$$\overline{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 3 & 1 & 1 & 5 \\ 1 & 4 & -1 & 4 \end{bmatrix} \overset{-3E_1+E_2\to E_2}{\Leftrightarrow} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & -2 & -4 \\ 1 & 4 & -1 & 4 \end{bmatrix} \overset{-E_1+E_3\to E_3}{\Leftrightarrow} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & -2 & -4 \\ 0 & 3 & -2 & 3 \end{bmatrix} \overset{\frac{3}{2}E_2+E_3\to E_3}{\Leftrightarrow} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & -2 & -4 \\ 0 & 0 & -10 & -10 \end{bmatrix}$$

$$-\frac{a_{j1}}{a_{11}}E_1 + E_j \to E_j$$

$x_1 + x_2 + x_3 = 3 \Rightarrow x_1 = 3 - x_2 - x_3 = 3 - 1 - 1 = 1$

$-2x_2 - 2x_3 = -4 \Rightarrow x_2 = 2 - x_3 = 1$

$-10x_3 = -10 \Rightarrow x_3 = 1$

Eliminarea gaussiana

```
function x = Gepp(A,b)
    [~,n] = size(A);
    x=zeros(size(b));
    A=[A,b];
    for i = 1:n-1
        [u,p] = max(abs((A(i:n,i))));
        p = p + i - 1;
        if u == 0, error('no unique solution'), end
        if p ~= i
            A([i,p],i:n+1) = A([p,i],i:n+1);
        end
        j=i+1:n;
        m=A(j,i)/A(i,i);
        A(j,i+1:n+1) = A(j,i+1:n+1)-m*A(i,i+1:n+1);
    end
    x(n) = A(n,n+1)/A(n,n);
    for i=n-1:-1:1
        x(i) = (A(i,n+1)-A(i,i+1:n)*x(i+1:n))/A(i,i);
    end
end
```

```
function x = gausselim(A, b)

  A = [A b];
  n = length(b);

  for k = 1:n - 1

    [valmax, pozmax] = max(abs(A(k:end, k)));
    pozpivot = pozmax + k - 1;

    if valmax == 0

      disp("NU avem solutie unica");
      return;

    elseif pozpivot ~= k

       A([k, pozpivot], k:end) = A([pozpivot, k], k:end);

    end

    for i = k + 1:n

      A(i, k:end) = A(i, k:end) - A(k, k:end) * (A(i, k) / A(k, k));

    end

  end

  x = backsubst(A(:,1:n), A(:, n + 1));

end

gausselim([1,1,1;3,1,1;1, 4, -1],[3;5;4])
```

```
ans = 3×1
           1
           1
           1
```

```
Gepp([1,1,1;3,1,1;1, 4, -1],[3;5;4])
```

```
ans = 3×1
           1
           1
           1
```

descompunere LU a lui A : $A = \mathrm{LU}$

unde L este Δ inferior, cu 1 pe diagonala, iar U este Δsup

$\mathrm{Ax} = b \Rightarrow \mathrm{LUx} = b \Rightarrow \mathrm{Ly} = b \Rightarrow \overset{\mathrm{Ly}=b}{\mathrm{Ux}=y}$

Matrice de permutare :

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

```
P=[0,1,0;1,0,0;0,0,1]
```

```
P = 3×3
     0     1     0
     1     0     0
     0     0     1
```

```
A=[1,2,3;4,5,6;7,8,9]
```

```
A = 3×3
     1     2     3
     4     5     6
     7     8     9
```

Produsul A*P ⇒ **permutare de coloane**

```
A*P
```

```
ans = 3×3
     2     1     3
     5     4     6
     8     7     9
```

Produsul A*P ⇒ **permutare de linii**

```
P*A
```

```
ans = 3×3
     4     5     6
     1     2     3
     7     8     9
```

Descompunere LUP

(L,U,P), L Δ inferior cu 1 pe diagonala, U Δ superior, iar PA = LU

```
function [L,U,P] = lup(A)
    [m,n] = size(A);
    P=zeros(m,n);
    piv=(1:m)';
    for i = 1:m-1
        [~,kp] = max(abs(A(i:m,i)));
        kp = kp + i - 1;
        if i ~= kp
            A([i,kp],:) = A([kp,i],:);
            piv([i,kp]) = piv([kp,i]);
        end
        % Complementul Schur
```

```
        lin = i + 1:m;
        A(lin,i) = A(lin,i) / A(i,i);

        A(lin,lin) = A(lin,lin) - A(lin,i)*A(i,lin);
    end
    for i = 1:m
        P(i,piv(i)) = 1;
    end
    U=triu(A);
    L=tril(A,-1) + eye(m);
end

function x = lupsolve(A,b)
    [L,U,P] = lup(A);
    y=forwardsubst(L,P*b);
    x=backsubst(U,y);
end
```

Descompunere Cholesky

$A = R^* R$ sau $A = R^T R$

$\mathrm{Ax} = b \Leftrightarrow R^* \mathrm{Rx} = b \Rightarrow \begin{matrix} R^* y = b \\ \mathrm{Rx} = y \end{matrix}$

```
function R=Cholesky(A)
    [m,~] = size(A);
    for k=1:m
        if A(k,k) <=0
            error('matrix is not HPD');
        end
        for j=k+1:m
            A(j,j:m) = A(j,j:m) - A(k,j:m)*conj(A(k,j))/A(k,k);
        end
        A(k,k:m) = A(k,k:m)/sqrt(A(k,k));
    end
    R=triu(A);
end

function x = choleskysolve(A,b)
    R = Cholesky(A);
    R_star = R';
    y=forwardsubst(R_star,b);
    x=backsubst(R,y);
end
```

Descompunere QR

Fie sistemul Ax = b, A= R^{nxm}

A se descompune in produs de doua matrici: A=QR, unde Q este matrice ortogonala ($QQ^T = I$) iar R este Δ superior

Astfel sistemul devine QRx = b

Inmultind cu Q^T la stanga obtinem: Rx = Q^Tb

Notam Q^Tb cu y => Rx = y care se poate rezolva cu backward substitution

```
function [Q, R] = descompunereQR(A)
    [m, n] = size(A);
    Q = eye(m);
    R = A;

    for k = 1:min(n,m)

        x = R(k:m, k);

        v = x;
        v(1) = v(1) + sign(x(1)) * norm(x, 2);
        v = v / norm(v, 2);

        R(k:m, k:n) = R(k:m, k:n) - 2 * v * (v' * R(k:m, k:n));

        Q(:, k:m) = Q(:, k:m) - 2 * Q(:, k:m) * v * v';
    end

end


function x = qrsolve(A,b)
    [Q,R] = descompunereQR(A);
    y = Q'*b;
    x=backsubst(R,y);
end
```

Probleme

Problema 1

Eliminarea Gaussiana

Problema 2

Descompunere LUP; Rezolvarea unui sistem folosind LUP

Problema 3

```
function [A,b] = genereazaSistem(n)
    A = randi([1, 15], n, n);
```

```
    b = sum(A,2);
end

[A,b] = genereazaSistem(3)
```

```
A = 3×3
     1     6     3
    13     6     6
     6    11     5
b = 3×1
    10
    25
    22
```

```
Gepp(A,b)
```

```
ans = 3×1
     1
     1
     1
```

```
lupsolve(A,b)
```

```
ans = 3×1
     1
     1
     1
```

Problema 4

Descompunerea Cholesky; Rezolvarea unui sistem folosind descompunerea Cholesky

```
choleskysolve([2 -1 0; -1 2 -1; 0 -1 2],[1;0;1])
```

```
ans = 3×1
     1
     1
     1
```

Problema 5

```
A = [3,-13,9,3;-6,4,1,-18;6,-2,2,4;12,-8,6,10]
```

```
A = 4×4
     3   -13     9     3
    -6     4     1   -18
     6    -2     2     4
    12    -8     6    10
```

```
b = [-19;-34;16;26]
```

```
b = 4×1
   -19
   -34
    16
    26
```

```
Gepp(A,b)
```

```
ans = 4×1
            3
            1
           -2
            1
```

```
lupsolve(A,b)
```

```
ans = 4×1
            3
            1
           -2
            1
```

```
try
    choleskysolve(A,b)
catch E
disp(E.message)
end
```

```
matrix is not HPD
```

Problema 6

```
M = [60,30,20;30,20,15;20,15,12]
```

```
M = 3×3
    60    30    20
    30    20    15
    20    15    12
```

```
[L,U,~] = lup(M)
```

```
L = 3×3
            1            0            0
          0.5            1            0
      0.33333            1            1
U = 3×3
           60           30           20
            0            5            5
            0            0      0.33333
```

```
R = Cholesky(M)
```

```
R = 3×3
        7.746        3.873        2.582
            0       2.2361       2.2361
            0            0      0.57735
```

Problema 7

```
function [A,b] = obtineSistem(n)
    A = zeros(n) + eye(n) - tril(ones(n),-1);
    b = zeros(n,1);

    for i = 1:n
        A(i,n) = 1;
```

```
        b(i,1) = -i+2 + 1;
    end
end

[A,b] = obtineSistem(4)
```

```
A = 4×4
     1     0     0     1
    -1     1     0     1
    -1    -1     1     1
    -1    -1    -1     1
b = 4×1
     2
     1
     0
    -1
```

```
xlu = lupsolve(A,b)
```

```
xlu = 4×1
        0.875
         0.75
          0.5
        1.125
```

```
xqr = qrsolve(A,b)
```

```
xqr = 4×1
        0.875
         0.75
          0.5
        1.125
```

```
A*xlu
```

```
ans = 4×1
     2
     1
     0
    -1
```

```
A*xqr
```

```
ans = 4×1
            2
            1
   2.2204e-16
           -1
```

Probleme suplimentare

Problema 8

Descompunere LUP; Rezolvarea unui sistem folosind LUP

Problema 9

Descompunerea QR; Rezolvarea unui sistem prin descompunere QR